



MOIZ DAWOODI

O/A LEVEL PHYSICS

O' LEVEL PHYSICS

General Wave Properties

3.1

STUDENT NAME

 0315 0805775

PHYSICS WITH
MOIZ DAWOODI

SYLLABUS OUTLINE:

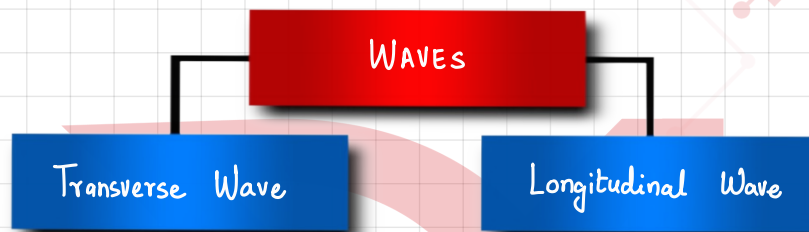
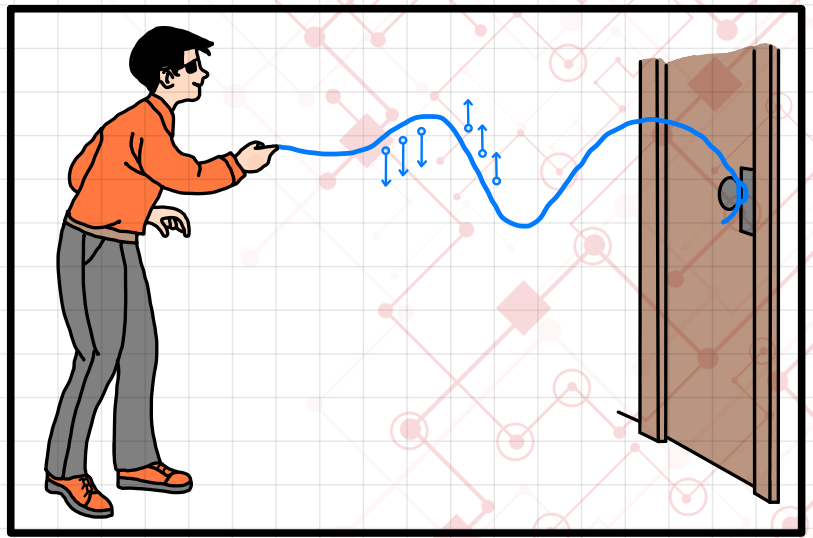
3.1 General properties of waves

- 1 Know that waves transfer energy without transferring matter
- 2 Describe what is meant by wave motion as illustrated by vibrations in ropes and springs and by experiments using water waves
- 3 Describe the features of a wave in terms of wavefront, wavelength, frequency, crest (peak), trough, amplitude and wave speed
- 4 Define the terms:
 - (a) frequency as the number of wavelengths that pass a point per unit time
 - (b) wavelength as the distance between two consecutive, identical points such as two consecutive crests
 - (c) amplitude as the maximum distance from the mean position
- 5 Recall and use the equation
wave speed = frequency $\times$ wavelength
 $v = f\lambda$
- 6 Know that for a transverse wave, the direction of vibration is at right angles to the direction of the energy transfer, and give examples such as electromagnetic radiation, waves on the surface of water, and seismic S-waves (secondary)
- 7 Know that for a longitudinal wave, the direction of vibration is parallel to the direction of the energy transfer, and give examples such as sound waves and seismic P-waves (primary)
- 8 Describe how waves can undergo:
 - (a) reflection at a plane surface
 - (b) refraction due to a change of speed
 - (c) diffraction through a gap
- 9 Describe how wavelength and gap size affects diffraction through a gap
- 10 Describe the use of a ripple tank to show:
 - (a) reflection at a plane surface
 - (b) refraction due to a change in speed caused by a change in depth
 - (c) diffraction due to a gap
 - (d) diffraction due to an edge
- 11 Describe how wavelength affects diffraction at an edge

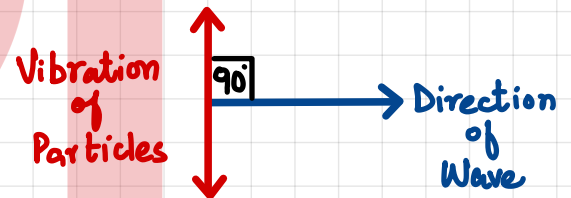
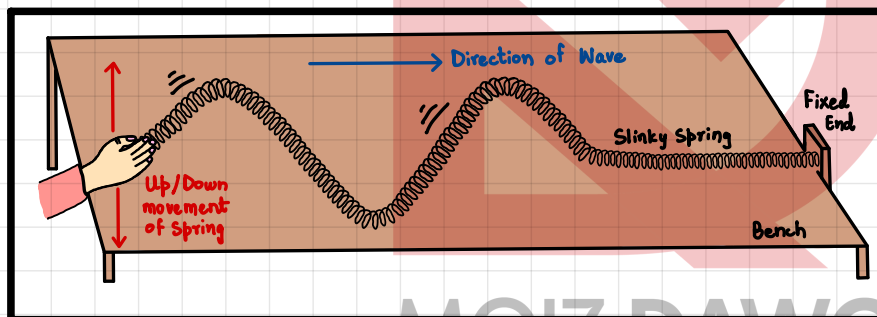


WAVE.

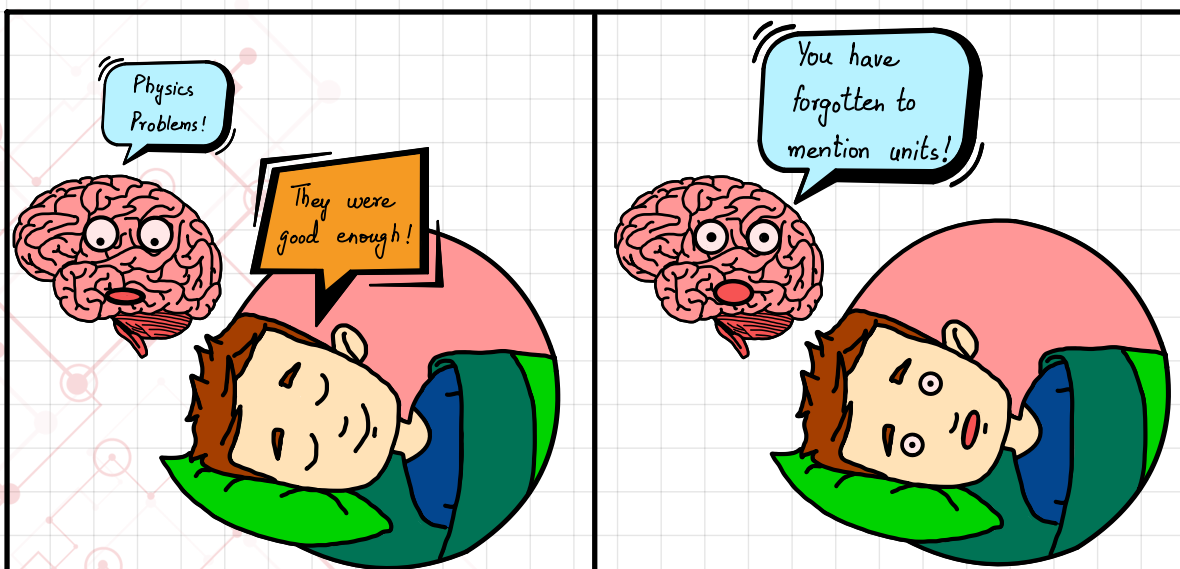
"Wave is the disturbance in the medium which transfers energy without transferring the medium."



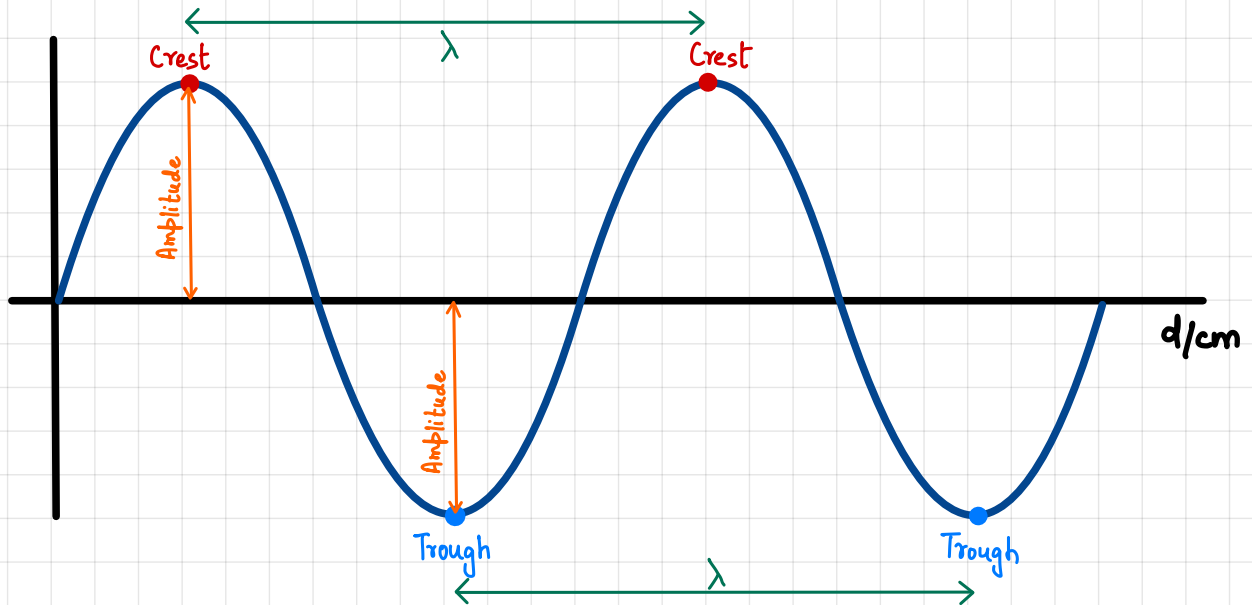
TRANSVERSE WAVE:



"A wave in which particles of the medium vibrate perpendicular to the direction of the wave is called Transverse Wave."



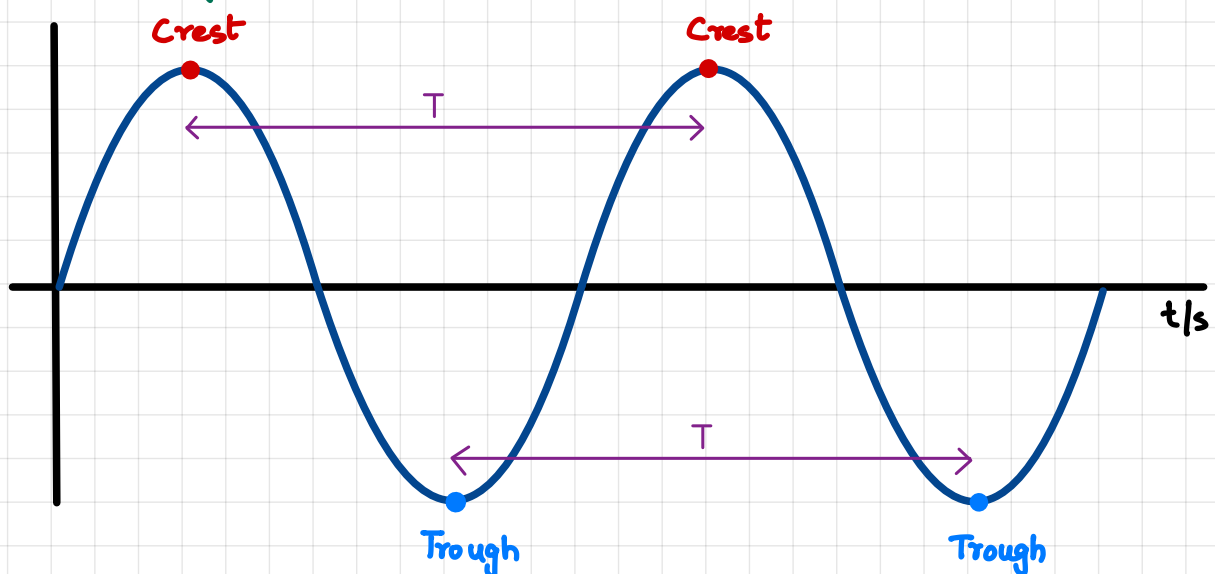
PROPERTIES OF WAVE MOTION:



- **CREST:** The maximum positive part of transverse wave.
- **TROUGH:** The maximum negative part of transverse wave.
- **AMPLITUDE:** The distance b/w mean to crest or mean to trough is called amplitude.

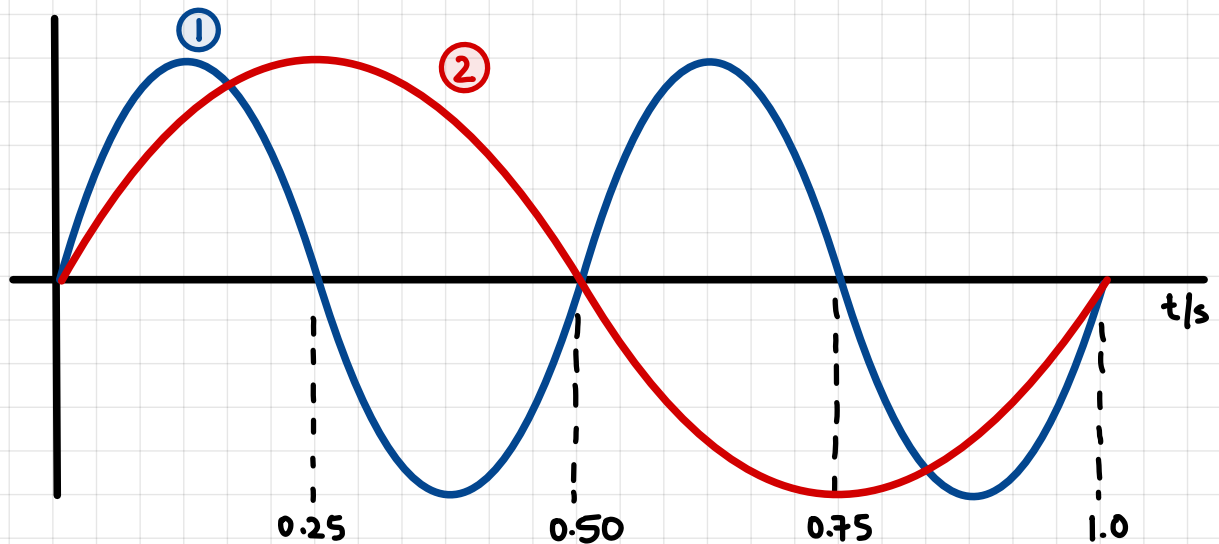
It is also known as the maximum displacement.

- **WAVELENGTH:** The distance b/w two consecutive crests or two consecutive troughs is called wavelength.
- It is denoted by Lambda (λ).



- **TIME PERIOD:** The time required to complete one wave is called Time Period or Period.
- **FREQUENCY:** No. of waves completed in a second is called frequency. Its S.I units is Hertz (Hz).

RELATIONSHIP BETWEEN TIME PERIOD & FREQUENCY:



WAVE # 01:

$$T = 0.5s$$

$$f = 2\text{ Hz}$$

WAVE # 02:

$$T = 1.0s$$

$$f = 1.0\text{ Hz}$$

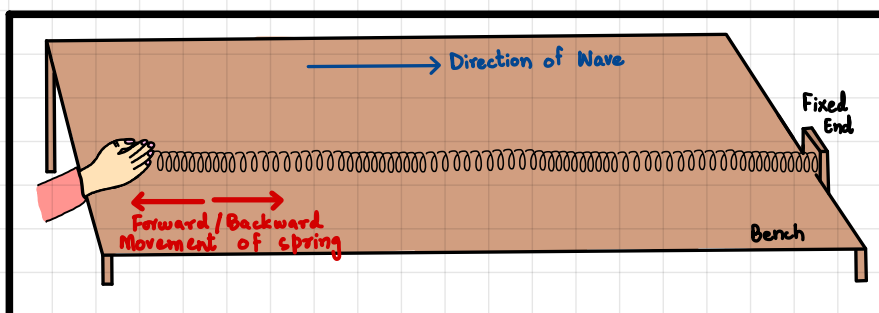
"When time period is doubled

$0.5s \rightarrow 1s$, Frequency halves $2\text{ Hz} \rightarrow 1\text{ Hz}$.

So time period is inversely proportional to the frequency.

$$f = \frac{1}{T}$$

LONGITUDINAL WAVE:



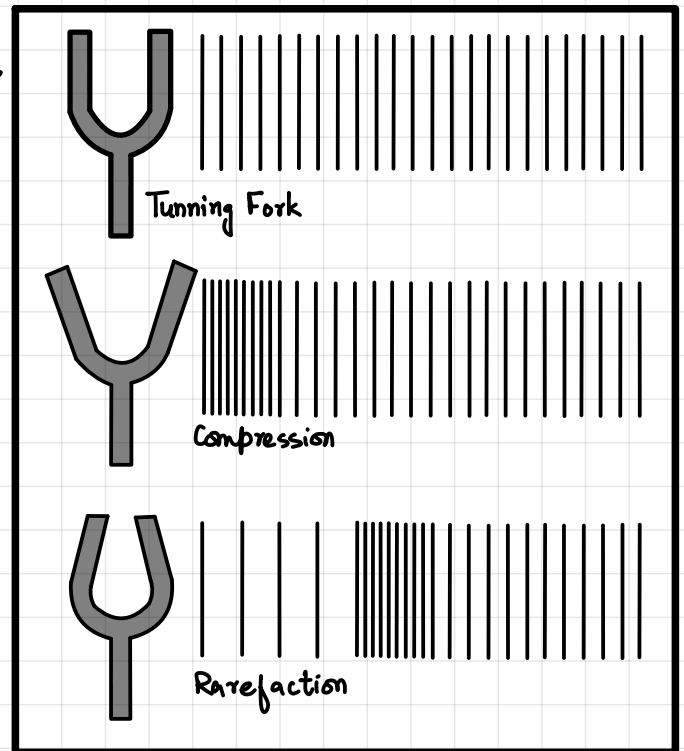
Vibration of Particles $\longleftrightarrow$ Direction of Wave

"A wave in which particles of the medium vibrate parallel to the direction of the wave is called Longitudinal Wave."

When tuning fork is hit with rubber its pronges vibrate back & forth.

When pronges move out, this creates the region of compression.

When pronges move in, this creates the region of rarefaction.



PROPERTIES OF WAVE MOTION:



● **COMPRESSION:** "The region of Longitudinal wave where particles are closer to each other & pressure is high is called compression."

● **RAREFACTION :** "The region of Longitudinal wave where particles are away from each other & pressure is low is called rarefaction."

● **WAVELENGTH:** "The distance b/w two consecutive compressions or rarefactions is called wavelength."

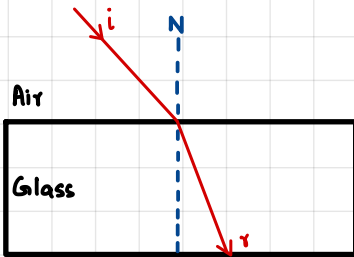
BACKGROUND CONCEPTS:

REFRACTION OF LIGHT:

"The bending of light due to change in medium is called refraction."

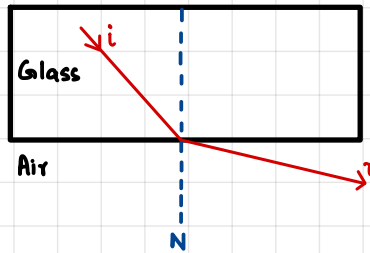
LESS DENSE $\rightarrow$ DENSER

- Light slows down.
- Moves towards normal.



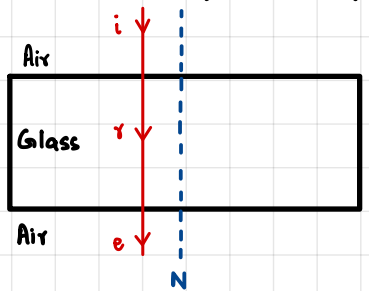
DENSER $\rightarrow$ LESS DENSE

- Light speeds up.
- Moves away from normal.

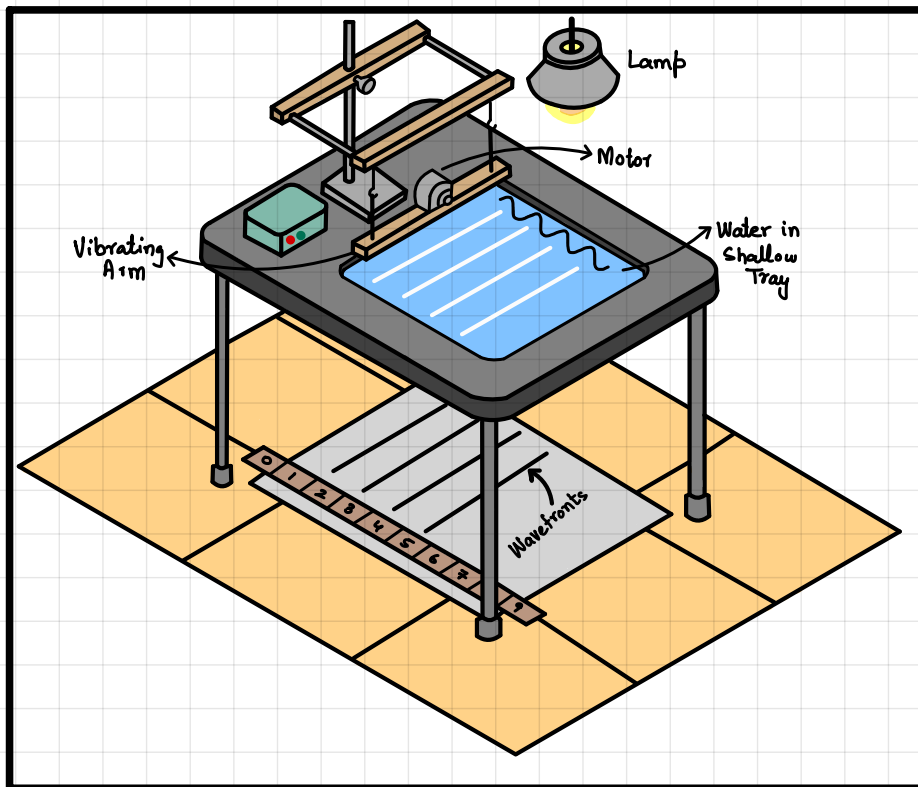


PARALLEL TO NORMAL

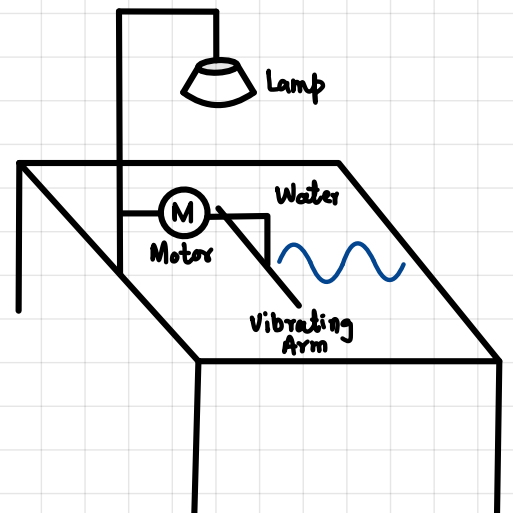
- Light doesn't bend.
- Slows down / speeds up.



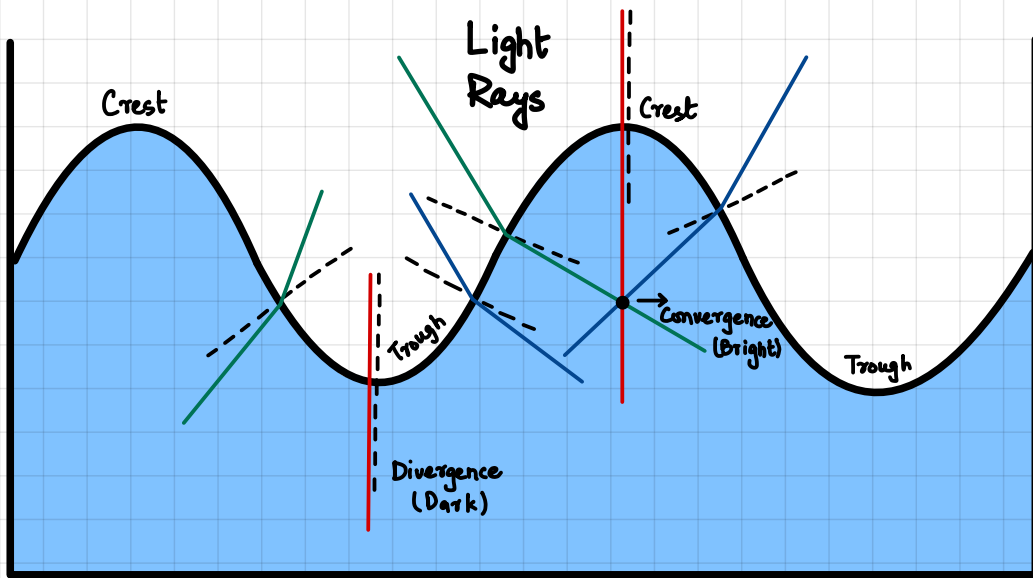
RIPPLE TANK:



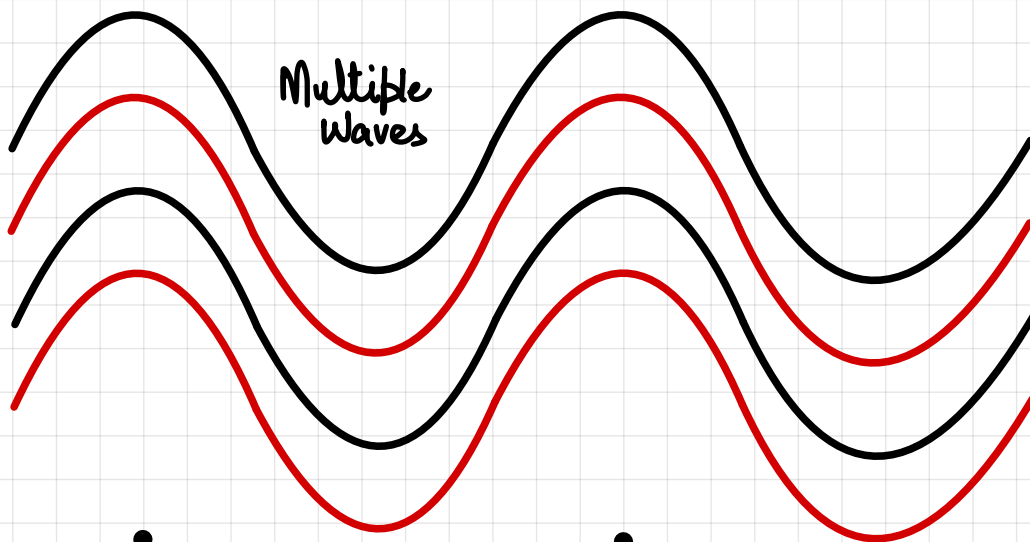
Draw a labelled diagram of ripple Tank.



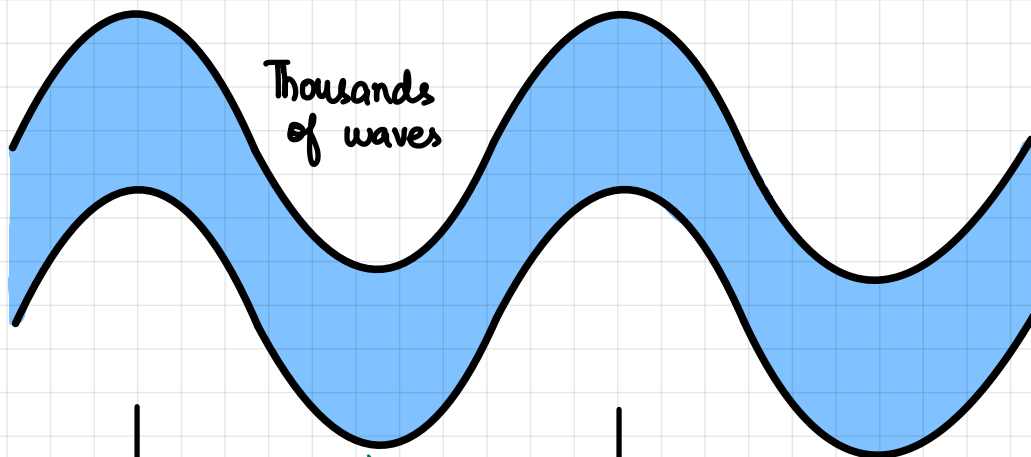
"Ripple Tank is a shallow glass tank of water used to demonstrate the basic properties of wave."



● → Bright spot made by Crest ← ●



●
● → Bright dots aligned ← ●
●
●



← →

"Bright line formed by all the crests aligned is called wavefront."

Wavefront Wavefront

WAVELENGTH: "The distance between two consecutive wavefronts is called wavelength."

WAVE EQUATION:

$$\text{Speed} = \frac{\text{Distance}}{\text{Time}}$$

$$v = \lambda \cdot \frac{1}{T}$$

$$v = f \lambda$$

Here frequency remains same.

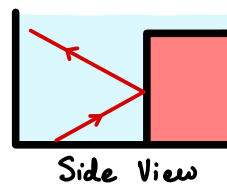
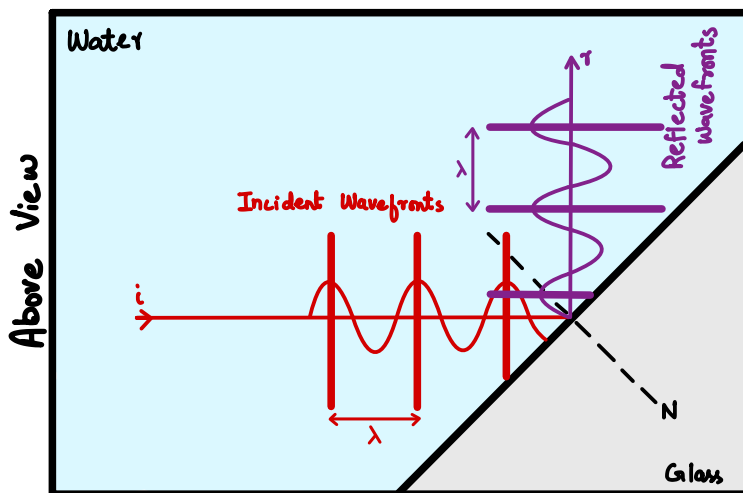
So,

$$v = \text{constant } \lambda$$

$$v \propto \lambda$$

REFLECTION OF WAVEFRONTS:

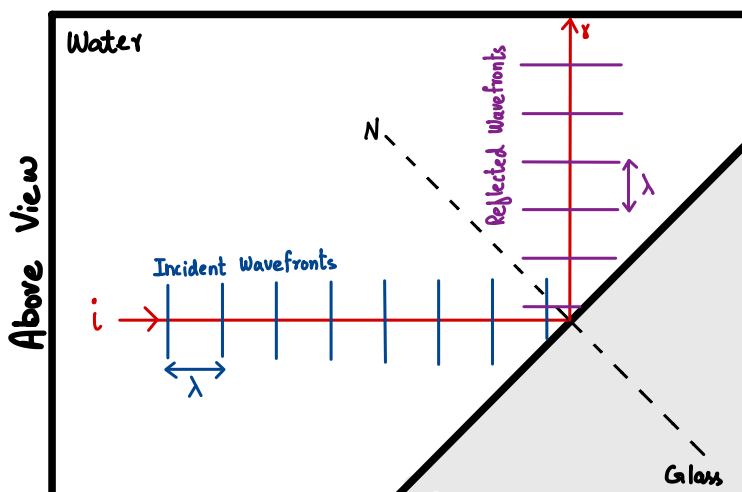
ROUGH DIAGRAM:



Laws of Reflection:

- Incident ray, reflected ray & normal lie on same plane.
- Angle of incidence is equal to angle of reflection.

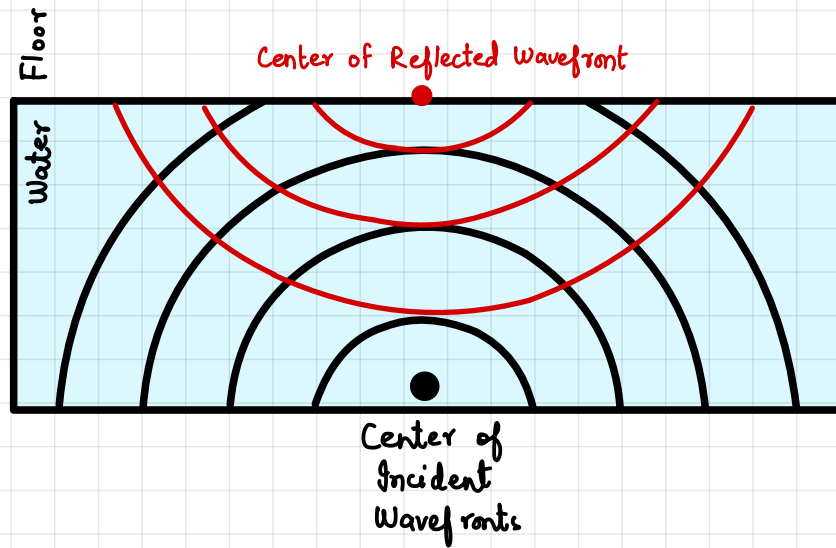
DRAW A LABELLED DIAGRAM OF REFLECTION OF PLANE WAVEFRONTS:



Here:

- Speed: Constant
- Wavelength: Constant
- Frequency: Constant

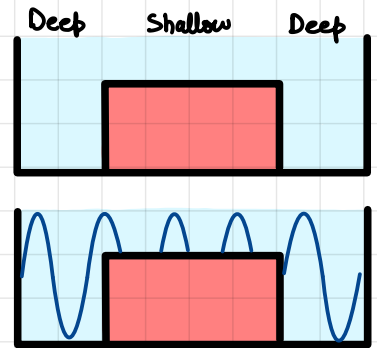
DRAW A LABELLED DIAGRAM OF REFLECTION OF CIRCULAR WAVEFRONTS:



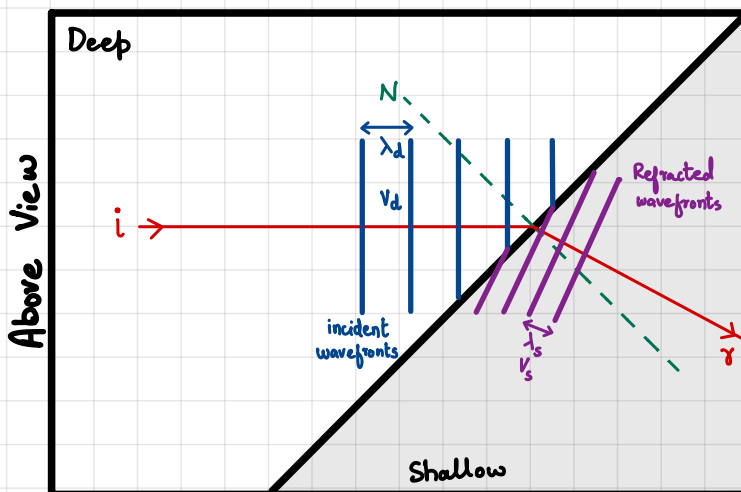
REFRACTION OF WAVEFRONTS:

Q Why wave dies out as it reaches the coastal area?

A Waves collide with sea bed due to less depth, loses its kinetic energy as heat & hence slows down.



DEEP TO SHALLOW:



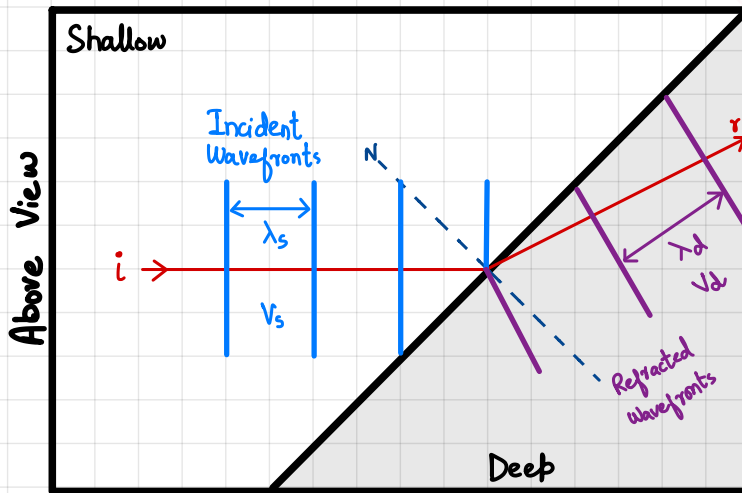
Here:

- Speed: Decrease
- Wavelength: Decrease ($v \propto \lambda$)
- Frequency: Constant

$$f_d = f_s$$

$$\frac{v_d}{\lambda_d} = \frac{v_s}{\lambda_s}$$

● SHALLOW TO DEEP:



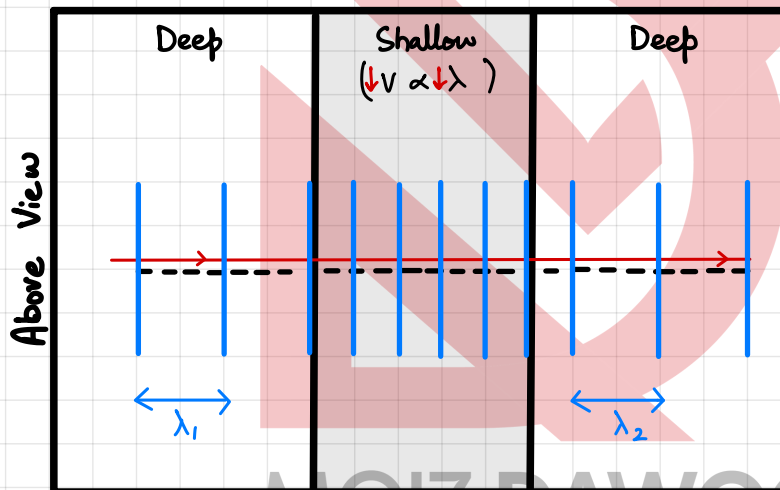
Here:

- Speed: **Increases**
- Wavelength: **Increases** ($v \propto \lambda$)
- Frequency: **Constant**

$$f_s = f_d$$

$$\frac{v_s}{\lambda_s} = \frac{v_d}{\lambda_d}$$

● PARALLEL TO THE NORMAL:



Deep to shallow, wave doesn't bend but slows down.

Shallow to deep, wave doesn't bend but speeds

up.

$$\lambda_1 = \lambda_2$$

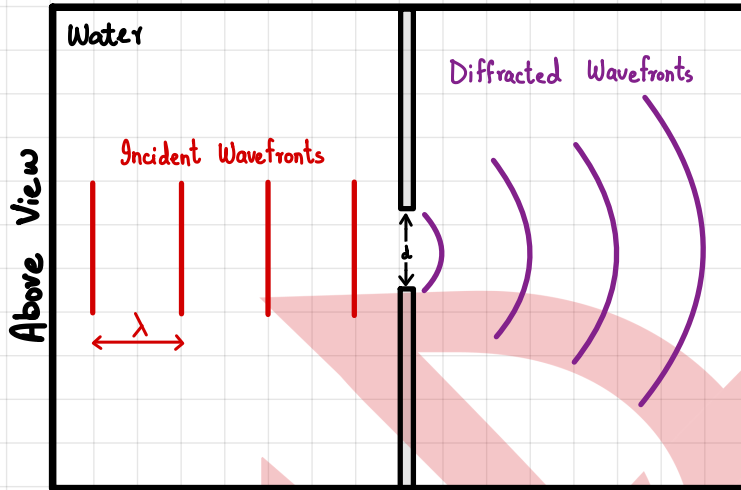
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DIFFRACTION

"The spreading of wave as it passes through narrow/wide gap or edges."

● DIFFRACTION THROUGH NARROW GAP: ($\lambda \geq d$, Circular wavefronts)

When gap width is less than or equal to wavelength, **circular wavefronts** are produced.

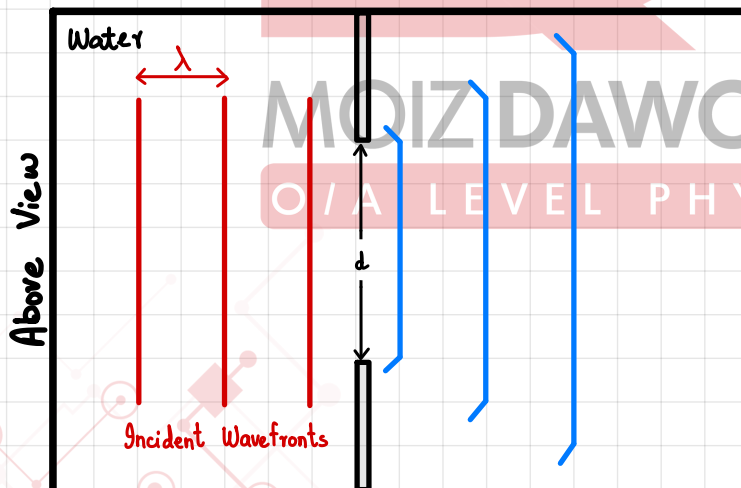


Here:

- Speed: Same
- Wavelength: Same
- Frequency: Same

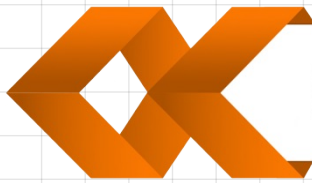
● DIFFRACTION THROUGH WIDE GAP & EDGES: ($\lambda < d$, Plane wavefronts)

When gap width is more than wavelength, plain wavefronts are produced though due to edges, some curvature is produced.



Here:

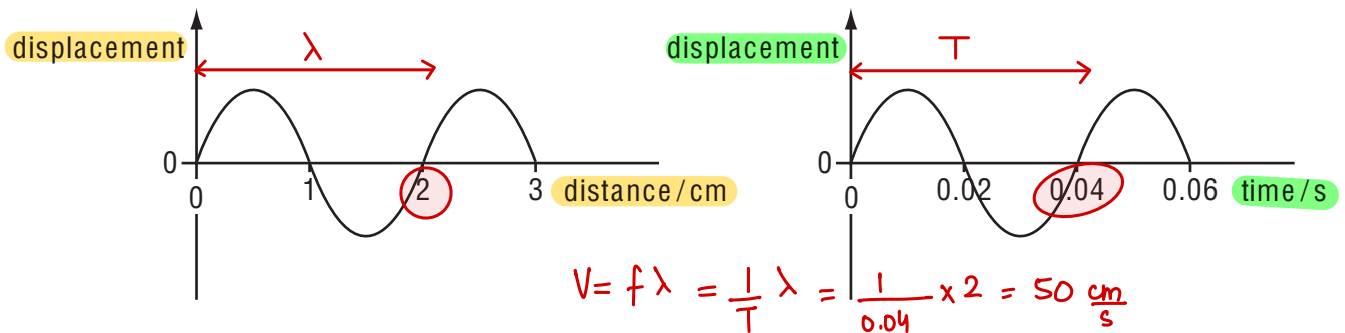
- Speed: Same
- Wavelength: Same
- Frequency: Same



MATCH THE FOLLOWING

Wave is the	particles vibrate perpendicular to the direction of wave.
Transverse wave makes	which demonstrates properties of waves.
Longitudinal wave makes	disturbance in the medium which transfer energy but no medium
Ripple tank is a device	particles vibrate parallel to the direction of wave.
Wavefront defines	reciprocal of frequency.
Frequency is the	a line joining all the crests.
Time period is the	spreading of wave through narrow opening or edge.
For ripple tank experiments	shallow region & wavelength decreases.
Diffraction defines	no. of waves completed in a second.
Waves slow does in	is the distance between two consecutive crests
Wavelength of transverse wave	frequency remains same.

- 01 The displacement-distance and displacement-time graphs are for a water wave in a ripple tank.



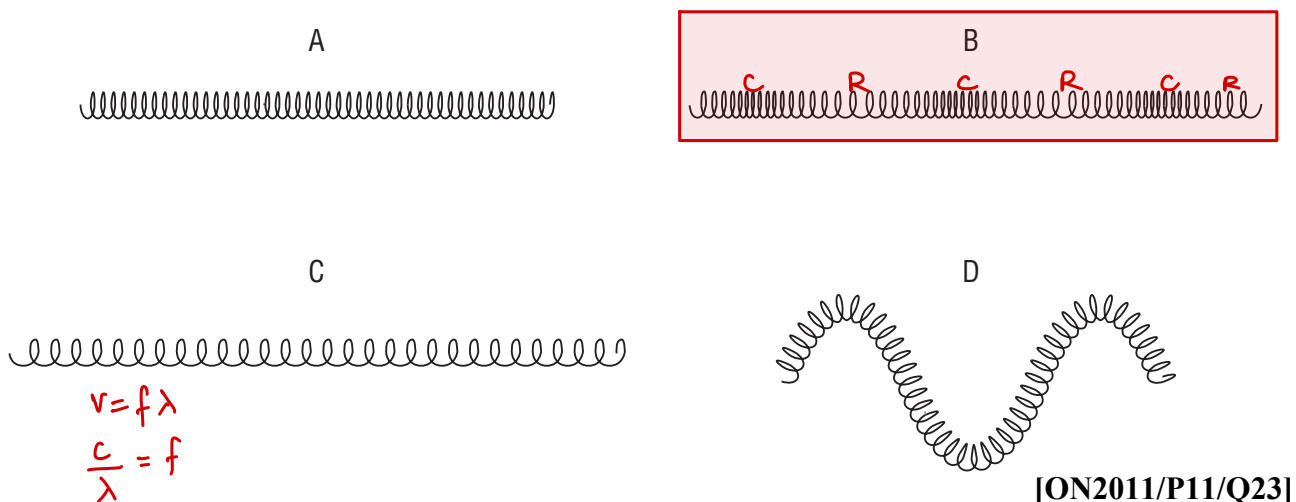
What is the speed of the water wave?

- A 0.02 cm/s B 0.08 cm/s C 25 cm/s **D 50 cm/s** [MJ2011/P11/Q20]

- 02 A student uses a spring to demonstrate waves. He moves the spring with his hand.



Which diagram demonstrates the type of wave produced by a source of sound?



- 03 The speed of radio waves is c . A radio station transmits waves with a wavelength λ .

What is the frequency of the transmission?

- A $\frac{c}{\lambda}$** B $\frac{\lambda}{c}$ C $\frac{1}{\lambda}$ D $c\lambda$ [MJ2012/P11/Q21]

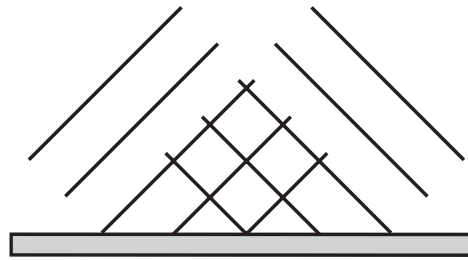
- 04 A wave has a frequency of 2 Hz.

How many waves are produced in one minute?

- A 2×60** B $\frac{60}{2}$ C 2 D $\frac{2}{60}$ [MJ2012/P12/Q21]
- $1\text{s} \rightarrow 2 \text{ waves}$
 $60\text{s} \rightarrow 2 \times 60 = 120$

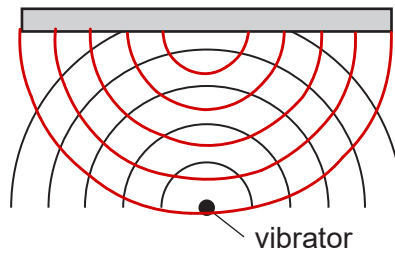
- 05 The diagram shows the pattern of waves in a ripple tank. What does the pattern show?

- A** waves being reflected
- B waves being refracted
- C waves changing frequency
- D waves changing speed



[ON2012/P11/Q21]

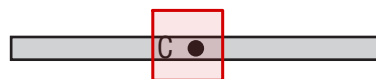
- 06 In a ripple tank, a vibrator produces circular wavefronts which hit a flat surface.



The reflected wavefronts are also parts of circles. Where is the centre of these circles?

A ●

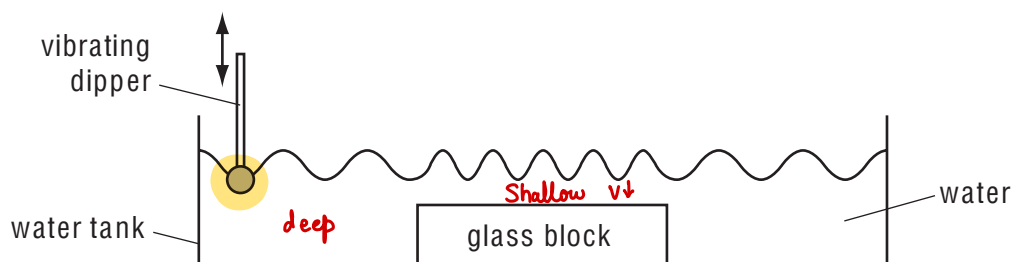
B ●



D ● vibrator

[ON2012/P12/Q21]

- 07 A ripple tank is used to show wave behaviour. The dipper vibrates up and down at a constant frequency.



What happens to the frequency and to the speed of the wave as it reaches the glass block?

	frequency	speed
A	decreases	decreases ✓
B	decreases	increases
C	remains the same ✓	decreases ✓
D	remains the same ✓	increases

[MJ2013/P11/Q17]

08 A wave has a frequency of 10 kHz. 10,000 Hz

Which pair of values of its speed and wavelength is possible?

	speed m/s	wavelength m
A	330	0.33
B	330	33
C	3.0×10^8	30
D	3.0×10^8	3.0×10^4

[MJ2013/P11/Q18]

09 Energy can be transferred in many different ways.

In which situation is energy transferred by wave motion?

- A colliding atoms in a heated copper rod
- B fast-moving electrons in a cathode-ray oscilloscope
- C hot water rising in a heated saucepan
- D ripples passing across water in a ripple tank

[MJ2013/P12/Q22]

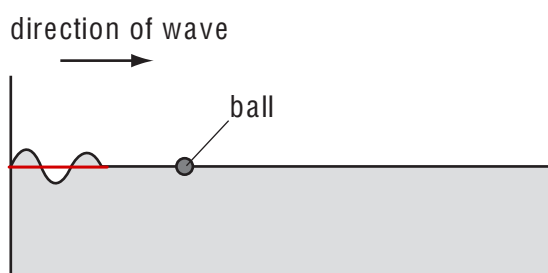
10 Which of the following travels as a longitudinal wave?

- A a radio wave in air
- B a sound wave in a solid
- C a wave on a rope shaken from side to side
- D an infra-red wave in space

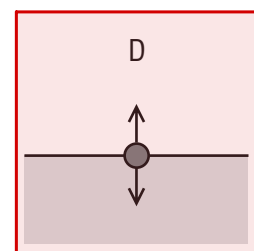
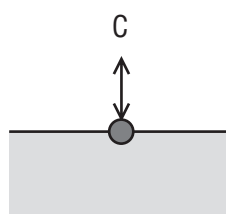
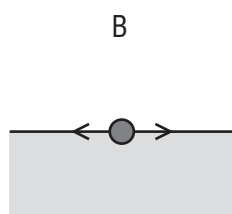
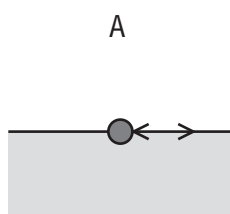
Transverse Wave
Light, Radio wave, Microwave, infrared, ultraviolet,
x-rays, gamma rays, water ripple.
Longitudinal Wave
Sound.

[MJ2013/P12/Q24]

11 The diagram shows a ball floating in a tank of water.



Which diagram shows the movement of the ball as the wave passes?

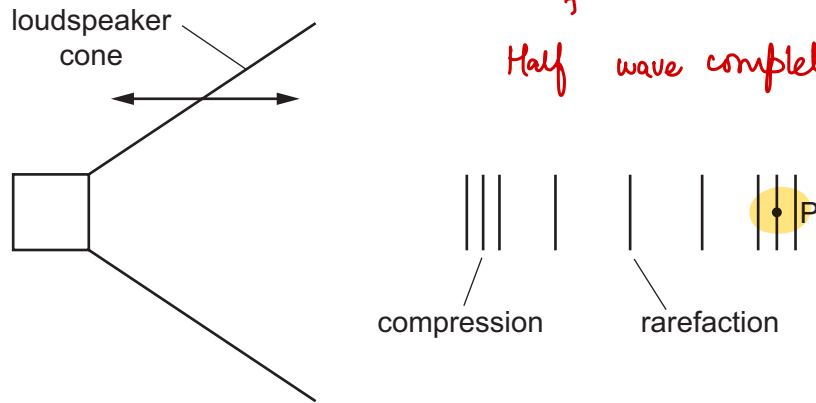


[ON2013/P11/Q19]

- 12 Compressions and rarefactions are sent out from a loudspeaker cone as it vibrates backwards and forwards. The frequency of vibration is 50 Hz.

$$T = \frac{1}{f} = 0.02 \text{ s}$$

Half wave completes in 0.01 s
 $\frac{0.02}{2}$ ↑



A compression is at point P. How much time elapses before the next rarefaction arrives at P?

- A 0.010 s B 0.020 s C 25 s D 50 s [MJ2014/P11/Q22]

- 13 Water waves refract at a boundary between deep water and shallow water.

What is the effect on the frequency, wavelength and speed of the waves at the boundary?

	frequency	wavelength	speed
A	changes	changes ✓	stays the same
B	changes	stays the same	stays the same
C	stays the same ✓	changes ✓	changes ✓
D	stays the same ✓	stays the same	changes ✓

[MJ2014/P11/Q23]

- 14 What is the frequency of a wave?

- A The number of waves passing a fixed point per second.
 B The number of peaks added to the number of troughs passing a fixed point per second.
 C The time taken for one wave to pass a fixed point.
 D The time taken for the displacement to change from maximum to minimum.

[MJ2014/P12/Q25]

- 15 A wave of frequency 13 000 Hz travels 1300 m in 4.0 s.

$$v = \frac{d}{t} = \frac{1300}{4} = 325 \frac{\text{m}}{\text{s}}$$

What is the wavelength of the wave?

$$v = f\lambda, \quad \lambda = \frac{v}{f} = \frac{325}{13000} = 0.025 \text{ m}$$

- A 0.025 m B 0.40 m C 2.5 m D 40 m [ON2014/P11/Q19]

- 16 A wave in a ripple tank passes from a deeper to a shallower region and refracts.

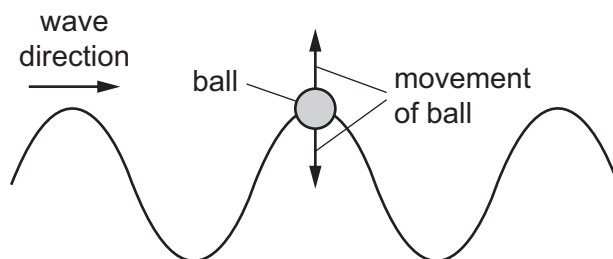
Which wave properties decrease as the wave enters the shallow region?

- A frequency only
 B speed only
 C frequency and wavelength
 D speed and wavelength

speed & wavelength decreases

[MJ2015/P11/Q21]

- 17 A ball floating in a ripple tank begins to move vertically up and down as a wave passes beneath it. The ball does not move horizontally.

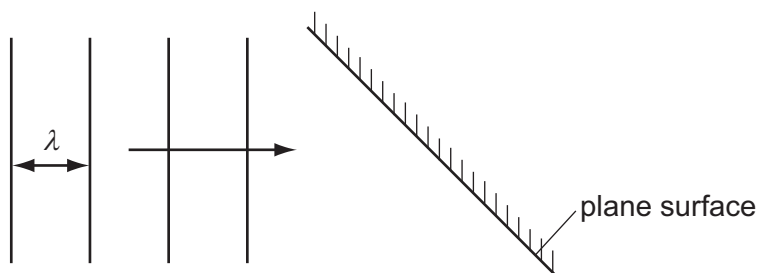


Which statement is correct?

- A Both energy and water are transferred in the wave direction.
- B Energy is not transferred in the wave direction but water is.
- C Energy is transferred in the wave direction but water is not.**
- D Neither energy nor water is transferred in the wave direction.

[MJ2015/P12/Q24]

- 18 In an experiment using a ripple tank, plane wavefronts arrive at a plane surface.



Which row correctly describes the waves after they are reflected from the surface?

	speed of waves	wavelength λ
A	larger	shorter
B	smaller	shorter
C	the same ✓	longer
D	the same ✓	the same ✓

[ON2015/P11/Q19]

P1

Mark Scheme

ANSWER KEY			
1	D	11	D
2	B	12	A
3	A	13	C
4	A	14	A
5	A	15	A
6	C	16	D
7	C	17	C
8	D	18	D
9	D		
10	B		

- 1 Fig. 1.1 shows circular wavefronts produced at the centre of a circular ripple tank.

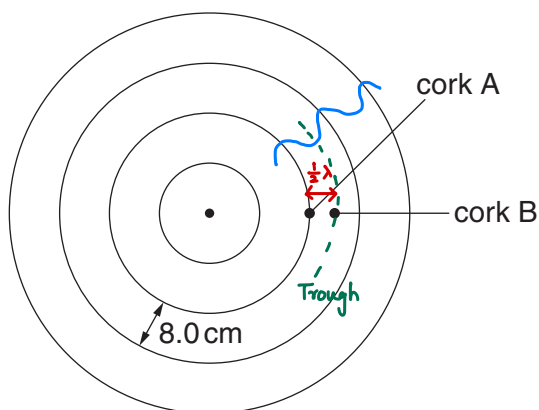


Fig. 1.1

Two corks, A and B, float on the water in the ripple tank. They move up and down on the surface of the water as the wave passes. The wavelength of the wave is 8.0 cm.

Fig. 1.2 shows how the displacement of A varies with time.

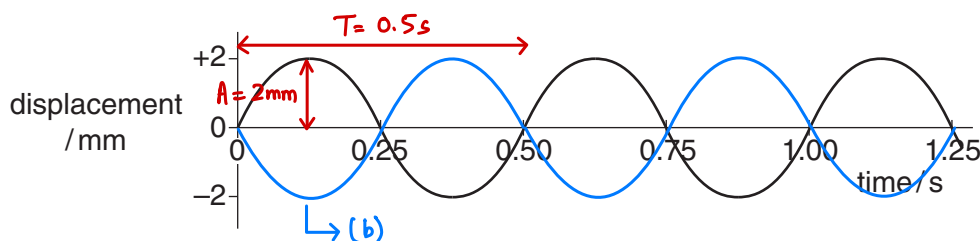


Fig. 1.2

- (a) State the amplitude of the vibrations of A as the wave passes.

amplitude = 2 mm [1]

- (b) The horizontal distance between A and B is half the wavelength of the wave.

On Fig. 1.2, sketch a graph to show how the displacement of B varies with time. [2]

- (c) (i) Use Fig. 1.2 to determine the frequency of the wave.

$$f = \frac{1}{T} = \frac{1}{0.5}$$

frequency = 2 Hz [2]



- (ii) The distance between the centre of the ripple tank and its edge is 40 cm.

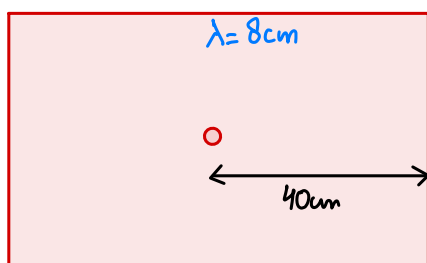
Determine the time taken by a wavefront to travel from the centre of the tank to the edge.

$$v = f \lambda = 2 \times 8 = 16 \text{ cm/s}$$

$$v = \frac{d}{t}$$

$$t = \frac{d}{v} = \frac{40}{16}$$

time = 2.5 s [2]



- 2 Fig. 2.1 shows a wave on the surface of water. The wave is travelling to the right.

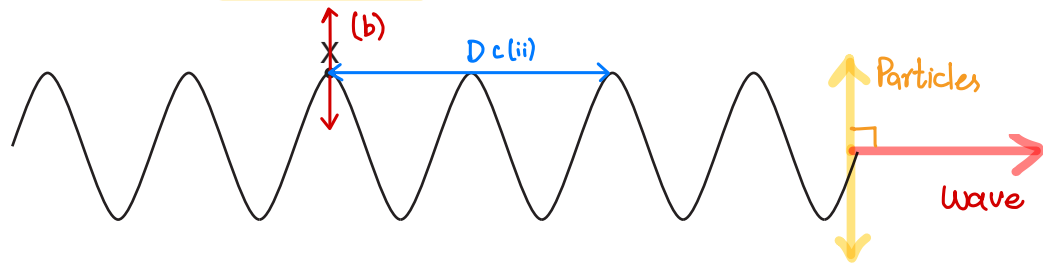


Fig. 2.1 (not to scale)

- (a) Describe what is meant by *wave motion*.

Wave motion is the disturbance in the medium, which transfers energy but don't transfer medium.
.....[2]

- (b) On Fig. 2.1, draw an arrow to show the direction of the movement of a water molecule at X. [1]

- (c) The frequency of the water wave is 2.0 Hz and the wavelength is 2.5 cm.

- (i) Calculate the speed of the wave.

$$v = f \lambda = 2 \times 2.5$$

speed = 5 cm/s[2]

- (ii) On Fig. 2.1, mark a distance which shows how far a wavefront at X moves in 1.0 s. Label this distance D. [1]

$$T = \frac{1}{f} = \frac{1}{2} = 0.5 \text{ s}$$

[MJ2012/P22/Q6]

- 3 A water wave of frequency 3.6 Hz is generated in a ripple tank. Fig. 3.1 is a scale diagram that shows the wavefronts travelling from deep water to shallow water in the ripple tank. A length of 1.0 cm on the diagram represents 8.0 cm in the ripple tank.

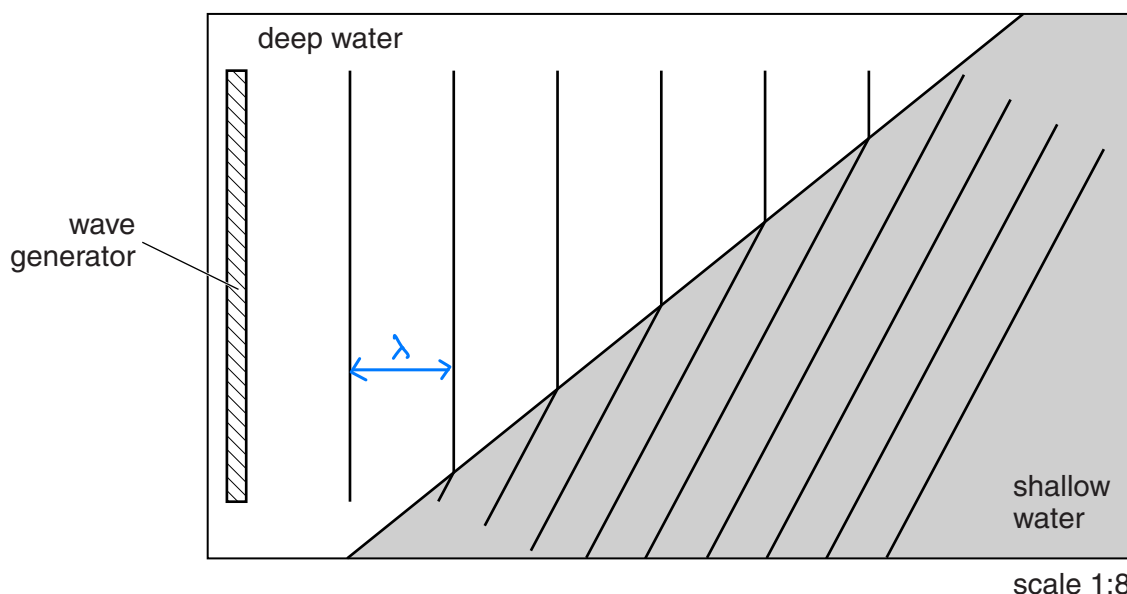


Fig. 3.1

- (a) (i) Using measurements from Fig. 3.1, determine the wavelength of the wave in the deep water.

$$\lambda = 1.2$$

$$\lambda_{\text{Actual}} = 1.2 \times 8$$

$$1 \text{ cm} \rightarrow 8 \text{ cm}$$

$$1.2 \text{ cm} \rightarrow \lambda_{\text{Actual}}$$

wavelength = 9.6 cm [2]

- (ii) Calculate the speed of the wave in the deep water.

$$v = f\lambda = 3.6 \times 9.6 = 34.56$$

speed = 34.56 cm/s [2]

- (b) As the wave passes into the shallow water, state what happens to

- (i) the frequency,

Remains same. [1]

- (ii) the speed.

Decreases. [1]

- 4 Fig. 4.1 shows part of a long rope used by a student to show a transverse wave.

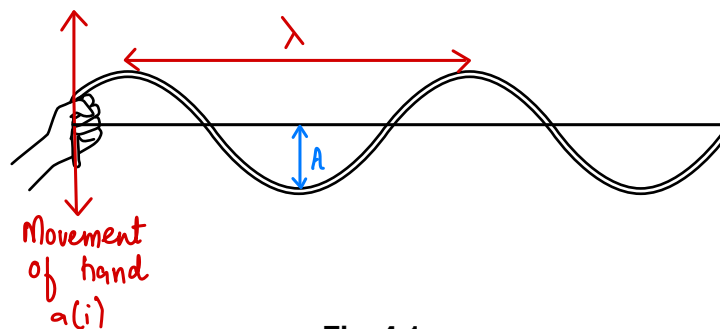


Fig. 4.1

- (a) On Fig. 4.1,

(i) mark the direction of movement of the student's hand, [1]

(ii) mark and label the wavelength λ of the wave, [1]

(iii) mark and label the amplitude A of the wave. [1]

- (b) Describe how the frequency of the wave is found using a stopwatch.

Measure time required to complete 10 waves using stopwatch then divide total time by 10 to get average time period (T). using formula $f = \frac{1}{T}$, frequency is found.

[2]

- (c) Using the same rope, the student produces a wave of a longer wavelength than that shown in Fig. 4.1.

$$v = f\lambda \quad f = \frac{v}{\lambda} \rightarrow \downarrow f \propto \frac{1}{\uparrow \lambda}$$

State how the student does this.

Decrease frequency.

[1]

[MJ2014/P22/Q4]

5 Ripple tanks are often used to illustrate wave motion.

(a) Describe what is meant by *wave motion* in a ripple tank.

.....
.....
.....[2]

(b) When describing wave motion, state what is meant by

(i) *frequency*,

.....[1]

(ii) *wavelength*.

.....
.....[2]

(c) A water wave in a ripple tank strikes a barrier. Fig. 5.1 shows some wavefronts of the incident wave.

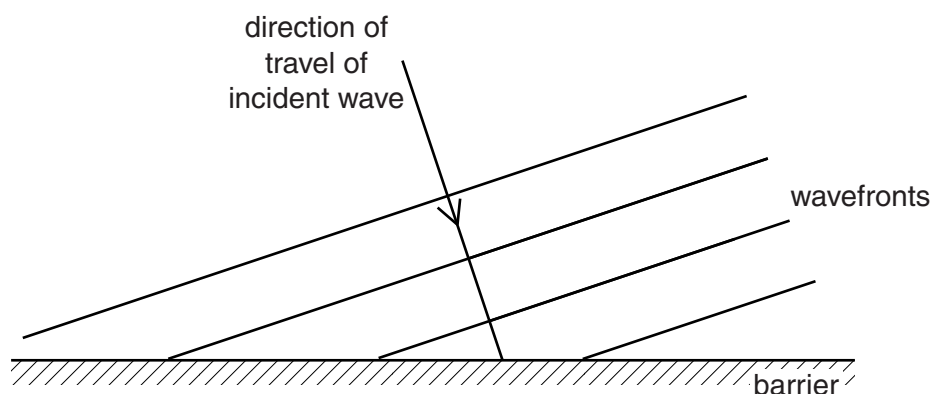


Fig. 5.1

The water wave hits the barrier and is reflected. Three of the wavefronts in Fig. 4.1 have already hit the barrier. The reflected parts of these wavefronts are not shown.

On Fig. 5.1, draw the reflected parts of these three wavefronts.

[3]

[ON2014/P21/Q4]

- 6 Fig. 6.1 shows a wave on a string. The wave is travelling towards the right.

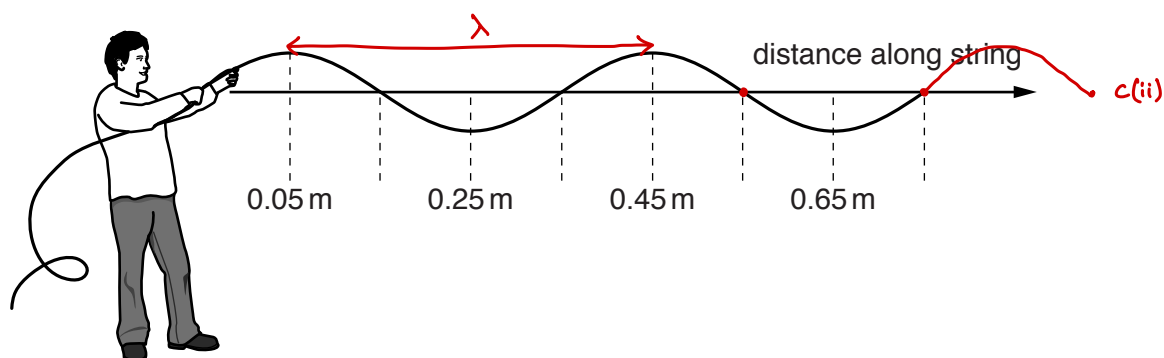


Fig. 6.1

- (a) Describe the movement of the particles of the string.

Particles are vibrating up & down perpendicular to the movement of wave on string.

.....[2]

- (b) Determine the wavelength of the wave.

$$\lambda = 0.45 - 0.05$$

wavelength = 0.40 m[1]

- (c) The speed of the wave is 2.0 m/s.

- (i) Calculate the frequency of the wave.

$$v = f \lambda$$

$$f = \frac{v}{\lambda} = \frac{2}{0.4}$$

frequency = 5 Hz[2]

- (ii) On Fig. 4.1, draw the string at a time 0.10 s later than the time in Fig. 6.1. [1]

$$T = \frac{1}{f} = \frac{1}{5} = 0.2 \text{ s}$$

[MJ2015/P21/Q4]

7 Fig. 7.1 shows the wavefronts of a water wave in deep water in a ripple tank.

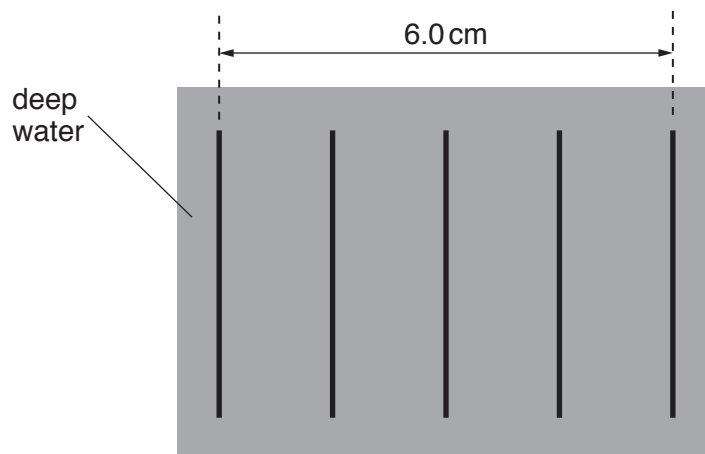


Fig. 7.1

The frequency of the wave in deep water is 5.0 Hz.

(a) Explain what is meant by the *frequency* of a wave.

.....

 [2]

(b) (i) Determine the wavelength of the wave in deep water.

..... [1]

(ii) Calculate the speed of the wave in deep water.

speed = [2]

(c) The wave passes from deep water into shallow water. The speed of the wave is less in shallower water.

(i) State and explain how this affects the wavelength of the wave.

.....

 [2]

- (ii) The wave in deep water shown in Fig. 7.2 travels towards the right and enters the shallow water at an angle. The wave refracts.

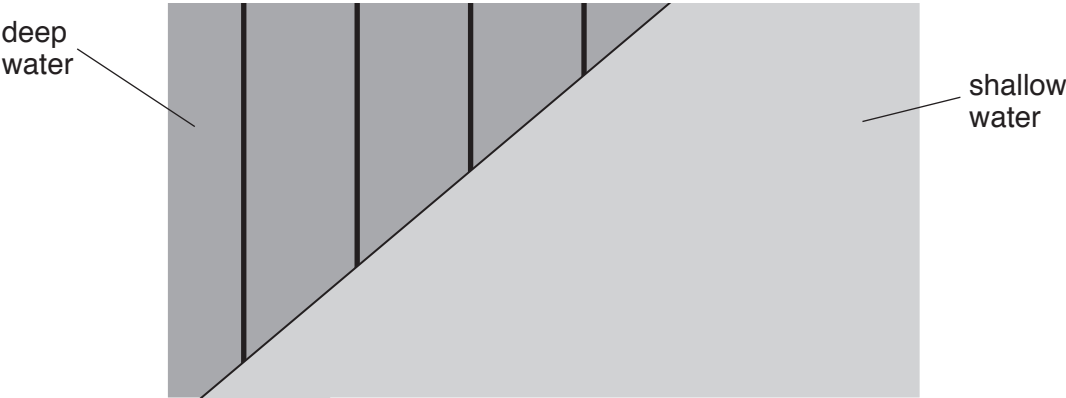


Fig. 7.2

On Fig. 7.2, draw the wavefronts in the shallow water. [3]

- (d) Sound is also a wave.
- (i) Describe one difference between a sound wave and a water wave.

.....

.....

.....

..... [2]

Page 3	Mark Scheme: Teachers' version	Syllabus	Paper
	GCE O LEVEL – May/June 2011	5054	22

- 4 (a) 2(.0) mm B1
 (b) same period (by eye), with at least one wave opposite phase to wave drawn B1
 (c) (i) ($f = 1/T$) numerical or algebraic seen (e.g. $1/0.5$) // 1 wave in 0.5 s // 2 waves in 1 s C1
 2(.0) Hz A1
 (ii) $v = f\lambda$ // 8×2 or $8 \times$ (i) // 16 (cm/s) // 5 (wavelengths from centre to edge) // ($t =$) d/v C1
 2.5 s **ecf** from (i) – i.e. **accept 5/(c)(i)** A1 [7]

Page 3	Mark Scheme: Teachers' version	Syllabus	Paper
	GCE O LEVEL – May/June 2012	5054	22

- 6 (a) Any 2 of
 • an oscillation/vibration/movement up and down
 • carries energy
 • no (net) movement of the medium/transfer of matter B2
 (b) arrow downwards or upwards or both B1
 (c) (i) ($v =$) $f\lambda$ in any form numerical or algebraic C1
 5(.0) cm/s or 0.05(0) m/s A1
 (ii) line or indication **labelled D** of length 2 wavelengths B1 [6]

Page 3	Mark Scheme	Syllabus	Paper
	GCE O LEVEL – October/November 2012	5054	21

- 5 (a) (i) clear attempt at measuring more than one wavelength e.g. 6.85/5 B1
 or 1.30 – 1.45 cm B1
 10.7 – 11.3 cm C1
 (ii) ($v =$) $f\lambda$ or $3.6 \times$ (a)(i) A1
 40(39.6) cm/s B1
 (b) (i) stays the same B1
 (ii) decreases B1 [6]

Page 3	Mark Scheme	Syllabus	Paper
	GCE O LEVEL – May/June 2014	5054	22

- 4 (a) (i) up and down clear, e.g. by double headed arrow **or** down [B1]
 (ii) any correct distance between consecutive points in phase [B1]
 (iii) correct distance [B1]
 (b) measure number of oscillations / count waves (passing) in a stated time or time at least one oscillation [B1]
 show how to calculate number of oscillations per second [B1]
 (c) **moves** (hand or rope) with slow(er) speed or rate / less frequency / less times per sec [B1]
 [6]

Page 2	Mark Scheme	Syllabus	Paper
	Cambridge O Level – October/November 2014	5054	21

- 4 (a) any **two** from:
 transmission of energy
 without net movement of medium
 through vibration of particles B2
 (b) (i) number of (complete) waves / cycles / oscillations per unit time / second B1
 (ii) distance between (neighbouring) waves C1
 distance between (neighbouring) wavefronts / points of same phase **or** crest to crest / trough to trough distance A1
 (c) three reflected wavefronts roughly correct direction M1
 wavelengths equal to each other and incident wavelength by eye A1
 reflected wavefronts joined to incident wavefronts B1 [8]

Page 2	Mark Scheme	Syllabus	Paper
	Cambridge O Level – May/June 2015	5054	21

- 4 (a) oscillate/vibrate stated or described B1
transverse movement described B1
(b) 0.40 m B1
(c) (i) $v = f\lambda$ or $(f =) v/\lambda$ or $2/(b)$ C1
5.0 Hz A1
(ii) clear attempt to draw wave moved along 0.20 m to right B1

Page 4	Mark Scheme	Syllabus	Paper
	Cambridge O Level – May/June 2015	5054	22

- 9 (a) number of waves (that pass a point) M1
or number of oscillations (passing a point)
in unit time or per second or in 1 second A1
(b) (i) 1.5 cm B1
(ii) $(v =)f\lambda$ or 5×1.5 seen C1
or $(s =)d/t$ and $f = 1/t$
7.5 cm/s A1
(c) (i) wavelength decreases B1
travels a shorter distance in the same time B1
or frequency stays the same (and $v = f\lambda$)
(ii) wavefronts with smaller wavelength (by eye) B1
smaller angle to surface (by eye) and slanted down B1
wavefronts join those in shallow water B1



xx✓



✓xx



xx✓



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✓xx



✓x✓



xx✓

(d) (i)

sound	water
particles / wave / source • vibrate / oscillate / move in direction of (travel of) wave / along wave • move backwards and forwards	particles / wave / source • vibrate / oscillate / move at 90° to direction of (travel of) wave • move up and down
(contains) compressions and rarefactions or particles come closer / further apart	(contains) crests and troughs
speed 300–330 m/s	wave slower (than sound)

B1
B1
one row only